$$\therefore \int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx .$$

$$\int_0^{\pi} g(\sin x)\,dx = 2\int_0^{\frac{\pi}{2}} g(\sin x)\,dx .$$

Result

$$\int_0^{\pi} g(\sin x)\,dx = 2\int_0^{\frac{\pi}{2}} g(\sin x)\,dx.$$

Note

The above result is useful in evaluating definite integrals of the type $\int_0^{\pi} g(\sin x)\,dx$.

Example 9.21

Evaluate $\int_0^{\pi} \frac{x}{1+\sin x}\,dx$.

Solution

$$\text{Let } I = \int_0^{\pi} \frac{x}{1+\sin x}\,dx .$$

$$= \int_0^{\pi} x\,\frac{1}{1+\sin x}\,dx$$

$$\text{Let } f(x) = \frac{1}{1+\sin x}. \text{ Then } f(\pi - x) = \frac{1}{1+\sin(\pi - x)} = \frac{1}{1+\sin x} = f(x)$$

$$\therefore \int_0^{\pi} \frac{x}{1+\sin x}\,dx = \frac{\pi}{2}\int_0^{\pi} \frac{1}{1+\sin x}\,dx, \quad \left(\because \int_0^{a} x\,f(x)\,dx = \frac{a}{2}\int_0^{a} f(x)\,dx \text{ if } f(a-x) = f(x)\right)$$

$$= \pi\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin x}\,dx, \quad \text{since } \int_0^{\pi} g(\sin x)\,dx = 2\int_0^{\frac{\pi}{2}} g(\sin x)\,dx$$

$$= \pi\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin\left(\frac{\pi}{2}-x\right)}\,dx \quad \text{since} \int_0^{a} f(x)\,dx = \int_0^{a} f(a-x)\,dx$$

$$= \pi\int_0^{\frac{\pi}{2}} \frac{1}{1+\cos x}\,dx = \pi\int_0^{\frac{\pi}{2}} \frac{1}{2\cos^2\frac{x}{2}}\,dx = \frac{\pi}{2}\int_0^{\frac{\pi}{2}} \sec^2\frac{x}{2}\,dx$$

$$= \pi\left[\tan\frac{x}{2}\right]_0^{\frac{\pi}{2}} = \pi\left[\tan\frac{\pi}{4} - \tan 0\right] = \pi .$$

Example 9.22

Show that $\int_0^{2\pi} g(\cos x)\,dx = 2\int_0^{\pi} g(\cos x)\,dx$, where $g(\cos x)$ is a function of $\cos x$.

Solution

Take $2a = 2\pi$ and $f(x) = g(\cos x)$.

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